

Spring 2025. Physics 510.

Quantum Theory II.

Test # 2.

Show all your work step by step and write down your final answers explicitly. Plain final answers will not be graded. You are allowed to use your equation sheet. A4, one side, handwritten by you personally.

1. Consider a three-dimensional repulsive ($V_0 > 0$) square barrier of radius a .

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

A particle of energy $E = \frac{\hbar^2 k^2}{2m} < V_0$ is incident on the well.

- Write down the Schrödinger equation for an s-wave ($l = 0$ mode).
- Solve it directly for both regions (it is possible because $l = 0$). Do not forget that the asymptotically free solution must contain a partial phase shift.
- Derive the phase shift, δ_0 , for the mode by applying necessary boundary conditions.
- How does the phase shift behave if $V_0 \rightarrow \infty$?
- Derive the total cross-section of the mode, σ_0 , in the limit of zero energy.

2. Calculate the scattering amplitude and differential cross-section in the Born approximation for the same potential

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

1. a) answer: Since there is spherical symmetry we only care about the radial part. So:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2M}{\hbar^2} (E - V_0) R - \frac{l(l+1)}{r^2} R = 0 \quad \text{for } r \leq a$$

Is our Schrödinger equation. Since it is an S wave we know that $l=0$. We can also set $R = \frac{u}{r}$ to simplify the equation. So:

$$\left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \left(-\frac{u}{r^2} + \frac{\partial u}{\partial r} \cdot \frac{1}{r} \right) \right) + \frac{2M}{\hbar^2} (E - V_0) \frac{u}{r} \right] = 0$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{\partial u}{\partial r} \cdot \frac{1}{r} - u \right) + \frac{2M}{\hbar^2} (E - V_0) \frac{u}{r} = 0$$

$$\frac{\partial^2 u}{\partial r^2} + \frac{2M}{\hbar^2} (E - V_0) u = 0$$

$$\text{Setting } K_{in}^2 = \frac{2M}{\hbar^2} (E - V_0)$$

$$\frac{\partial^2 u}{\partial r^2} + K_{in}^2 u = 0$$

$K_{in} < 0$ since $V_0 > E$, so:

$$u_{\text{inside}} = A e^{-K_{in} r} + B e^{K_{in} r}$$

We want u finite at $r=0$ so:

$$u_{\text{inside}}(0) = 0 = A + B \Rightarrow A = -B$$

$$U_{\text{inside}} = A (e^{-K_{\text{in}} r} - e^{K_{\text{in}} r})$$

Now let us calculate outside, $r \geq a$:

We will get:

$$\nabla_r^2 U + \frac{2M}{\hbar^2} (E - V) U = 0$$

Since $V = 0$ for $r \geq a$:

$$\nabla_r^2 U + \frac{2ME}{\hbar^2} U = 0$$

$$K_{\text{out}}^2 = \frac{2ME}{\hbar^2}$$

$$\nabla_r^2 U + K_{\text{out}}^2 U = 0$$

$$U_{\text{outside}} = C \cos(K_{\text{out}} r) + D \sin(K_{\text{out}} r)$$

Let us set $C = \sin(\delta_0)$ and $D = \cos(\delta_0)$

$$U_{\text{outside}} = \sin(\delta_0) \cos(K_{\text{out}} r) + \cos(\delta_0) \sin(K_{\text{out}} r)$$

$$U_{\text{outside}} = \sin(K_{\text{out}} r + \delta_0)$$

~~All that is left is the "A" constant for U_{inside} , since this is a square barrier with we know that at the radius $U_{\text{inside}} = 0$, so:~~

2) Answer: We use the continuity equations to find A and δ_0 , so:

$$\begin{cases} U_{\text{inside}}(a) = U_{\text{outside}}(a) \\ U'_{\text{inside}}(a) = U'_{\text{outside}}(a) \end{cases} \Rightarrow \begin{cases} A(e^{-K_{\text{in}}a} - e^{K_{\text{in}}a}) = \sin(K_{\text{out}}a + \delta_0) \\ -K_{\text{in}}A(e^{-K_{\text{in}}a} - e^{K_{\text{in}}a}) = K_{\text{out}}\cos(K_{\text{out}}a + \delta_0) \end{cases}$$

$$\Rightarrow \begin{cases} -K_{\text{in}}\sin(K_{\text{out}}a + \delta_0) = K_{\text{out}}\cos(K_{\text{out}}a + \delta_0) \\ \tan(K_{\text{out}}a + \delta_0) = -\frac{K_{\text{out}}}{K_{\text{in}}} \end{cases}$$

$$\begin{cases} A = \frac{\sin(K_{\text{out}}a + \tan^{-1}(-\frac{K_{\text{out}}}{K_{\text{in}}}) - K_{\text{out}}a)}{e^{-K_{\text{in}}a} - e^{K_{\text{in}}a}} \\ \delta_0 = \tan^{-1}(-\frac{K_{\text{out}}}{K_{\text{in}}}) - K_{\text{out}}a \end{cases}$$

$$\boxed{\begin{cases} A = \sin(\tan^{-1}(-\frac{K_{\text{out}}}{K_{\text{in}}})) \\ \delta_0 = \tan^{-1}(-\frac{K_{\text{out}}}{K_{\text{in}}}) - K_{\text{out}}a \end{cases}}$$

3) Answer: * If $V_0 \rightarrow \infty$ that implies a strong potential which makes this a low scattering process, so as $V_0 \rightarrow \infty$ $\frac{K_{\text{out}}}{K_{\text{in}}} \ll 1$ so $\tan^{-1}(-\frac{K_{\text{out}}}{K_{\text{in}}}) \approx -\tan^{-1}(\frac{K_{\text{out}}}{K_{\text{in}}}) \approx -\frac{K_{\text{out}}}{K_{\text{in}}}$

So our phase shift will be: $\delta_0 \approx -\frac{K_{\text{out}}}{K_{\text{in}}} - K_{\text{out}}a$

~~Sout~~

$$\boxed{\delta_0 \approx \frac{k_{out} - k_{in} k_{out} a}{k_{in}}}$$

~~2) Answer: $\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$~~

~~Since we only have $l=0$:~~

~~$$\sigma_0 = \frac{4\pi}{k^2} \sin^2 \delta_0$$~~

~~$$\sigma_0 \neq \frac{4\pi}{k^2} \sin^2 \delta$$~~

2) Answer: we know that: $f(\theta) = \frac{1}{k_{out}} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} P_l(\cos \theta) \sin \delta_l$

where $f(\theta)$ is the scattering amplitude. Since we are only working with an S wave we have that:

$$f(\theta) = \frac{1}{k_{out}} e^{i\delta_0} P_0(\cos \theta) \sin \delta_0$$

$$P_0(\cos \theta) = 1$$

~~we~~ We know from probability current conservation theorem that $\sigma = |f(\theta)|^2 / \frac{1}{4}$ so after plugging our $f(\theta)$ in we get:

$$\sigma_0 = \frac{\sin^2 \delta_0}{K_{out}^2}, \text{ since } |e^{i\delta_0}|^2 = 1$$

We know for a solid object $\mathcal{R} = 4\pi$, so:

$$\sigma_0 = \frac{4\pi}{K_{out}^2} \sin^2 \delta_0$$

→ which for our case will be the total cross section

Since we are working on low energies:

$$\sigma_0 = \frac{4\pi}{K_{out}^2} \sin^2 \delta_0 \approx \frac{4\pi}{K_{out}^2} \delta_0 \approx \frac{4\pi}{K_{out}^2} \left(\frac{K_{out} - k_{in} a}{K_{in}} \right)$$

2. Answer: $f(\theta) = -\frac{\mu}{2\pi\hbar^2} \int_0^\infty \frac{\sin q r}{q} V(r) dr$

using born approximation

$$f(\theta) = -\frac{\mu}{2\pi\hbar^2 q} \left(\int_0^a V(r) r \sin q r dr + \int_a^\infty V(r) r \sin q r dr \right)$$

$$f(\theta) = -\frac{\mu}{2\pi\hbar^2 q} \left(V_0 \int_0^a r \sin q r dr + 0 \right)$$

	D	I
+	π	$\sin q r$
-	1	$\frac{\cos q r}{q}$
+	0	$\frac{\sin q r}{(q r)^2}$

$$f(\theta) = -\frac{\mu V_0}{2\pi\hbar^2 q} \left(-\frac{a \cos q a}{q a} + \frac{\sin q a}{q^2 a^2} \right)$$

$$f(\theta) = \frac{m V_0}{2\pi \hbar^2 q} \left(-\frac{\sin qa}{q^2 a^2} + \frac{\cos qa}{qa} \right)$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

$$\frac{d\sigma}{d\Omega} = \frac{m^2 V_0^2}{4\pi^2 \hbar^4 q^2} \left(\frac{\cos qa}{qa} - \frac{\sin qa}{q^2 a^2} \right)^2$$

where $q = 2k \sin \frac{\theta}{2}$